

Master's Comprehensive Exam

II

Applied (90 minutes)

March 6, 2004

Instructions: Answer all three questions. Open book and open notes.

1. In an experiment designed to study the effects of illumination level on task performance, subjects were required to insert a fine-tipped probe into the eyeholes of ten needles in rapid succession both for a low light level with a black background and a higher level with a white background. The data below show the time to complete the task (in seconds).

Subject	1	2	3	4	5	6	7	8	9
Black	25.85	28.84	32.05	25.74	20.89	41.05	25.01	24.96	27.47
White	18.23	20.84	22.96	19.68	19.50	24.98	16.61	16.07	24.59

Do the data indicate that the higher level of illumination yields a decrease of more than 5 in the true average task completion time? Conduct a hypothesis test to address this question. State precisely any assumptions you make. State your conclusion *precisely* in terms a non-statistician can understand.

2. Here is a short description of study reported in the *Journal of the American Medical Association* a few years ago:

Temporal patterns of cocaine use in pregnancy. Perinatal outcome

Seventy-five cocaine-using women enrolled in a comprehensive perinatal care program were divided into two groups: those who used cocaine in only the first trimester of pregnancy (group 1 [N = 23]) and those who used cocaine throughout pregnancy (group 2 [N = 52]). Perinatal outcomes of these pregnancies were compared with perinatal outcomes of a matched group of obstetric patients with no history or evidence of substance abuse (group 3 [N = 40]). Group 2 women had an increased rate of preterm delivery and low-birth-weight infants as well as an increased rate of intrauterine growth retardation. Group 1 women had rates of these complications similar to the drug-free group. Mean birth weight, length, and head circumference for term infants were reduced in only the group 2 infants. However, both groups of cocaine-exposed infants demonstrated significant impairment of orientation, motor, and state regulation behaviors on the Neonatal Behavioral Assessment Scale.

- a. Consider specifically the birthweight outcome. What sort of statistical analysis method would you use if you had to analyze the data from this study?
 - b. Assume that group 2 and group 3 had significantly different mean birthweights (p -value = 0.02). What conclusion can you draw?
 - c. Do you have statistical concerns about the study design? What questions would you like to ask the investigators?
3. Eighty-five food critics rated the food at three university dining halls, labeled A, B, and C, with the following results:

	A	B	C
Tasty	3	10	4
Not Tasty	8	19	41

- a. Perform Pearson's Chi-square test to test the null hypothesis that the proportion of raters rating food as "Tasty" is the same in all three dining halls. Report a p -value, and draw a conclusion. Assume that raters are assigned to dining halls at random.